Universität Konstanz

Applied Natural Language Processing (NLP) for Online Harm Detection

Jun.-Prof. Dr. Miriam Schirmer

Konstanz, 22.10.2026

Outline for Today

1. Topic Intro
2. Admin
3. Questions



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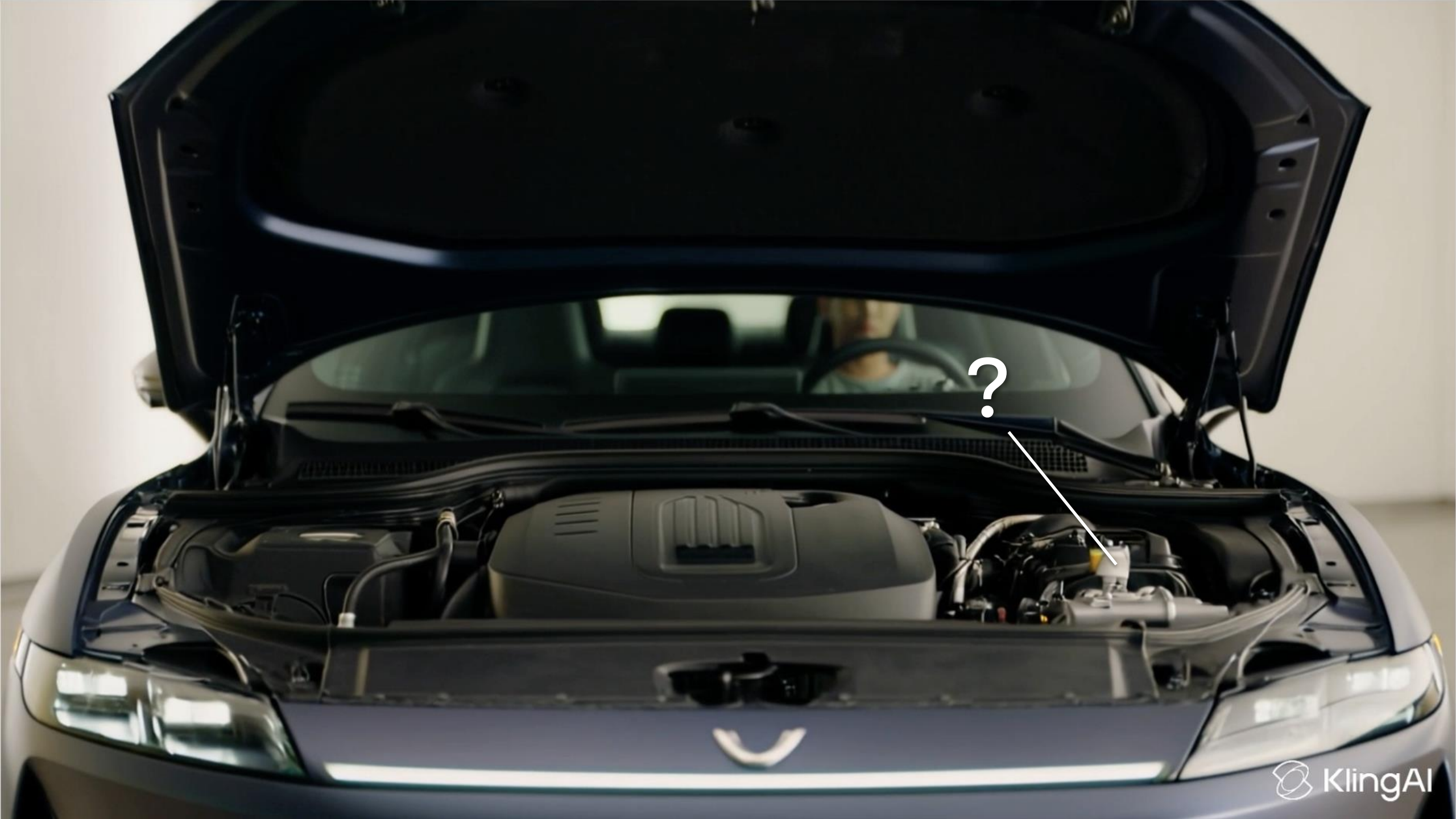


What comes to mind when you hear "language model"?

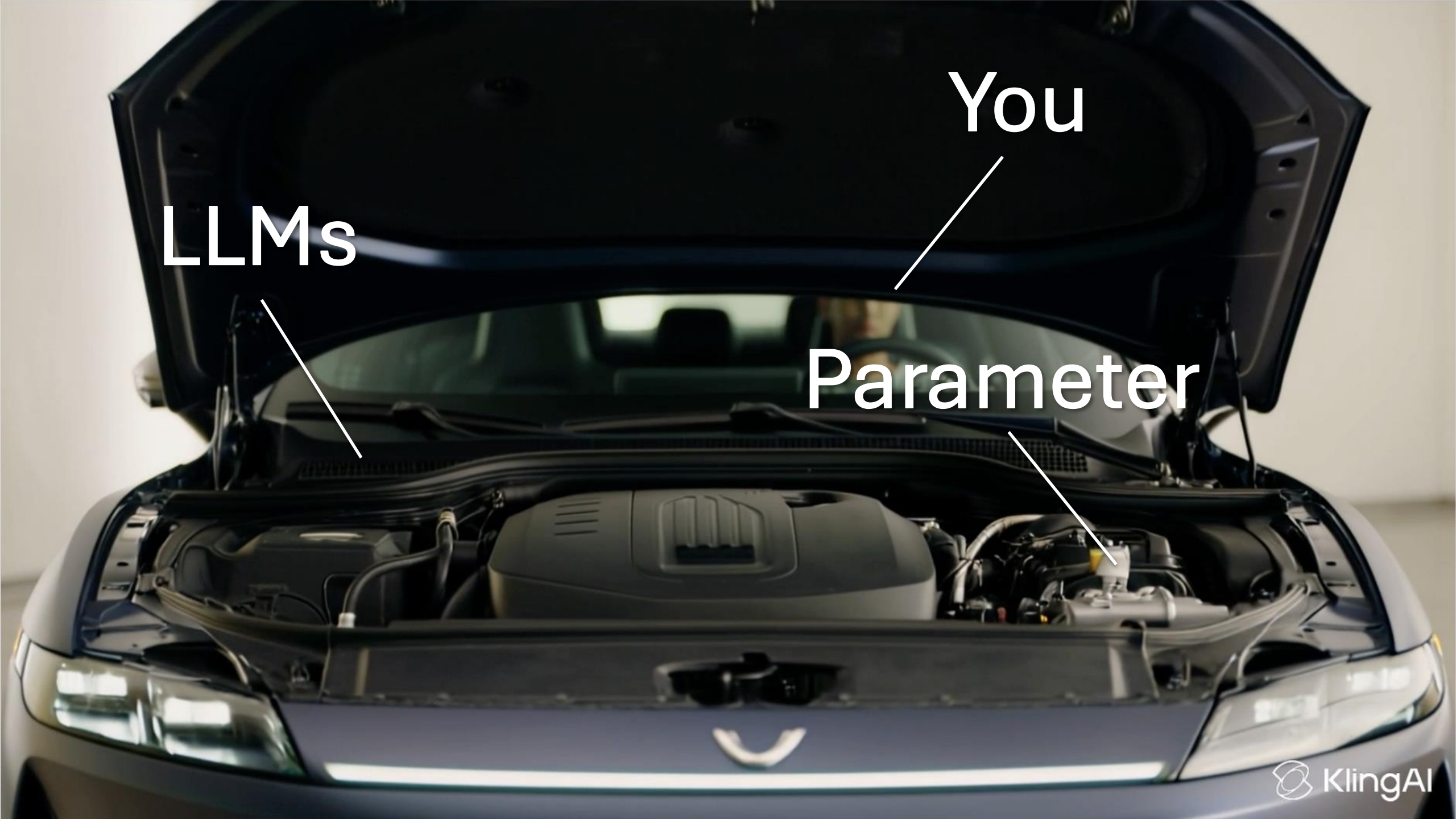
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KlingAI


?
KlingAI


Washer fluid reservoir
KlingAI


You
LLMs
Parameter
KlingAI

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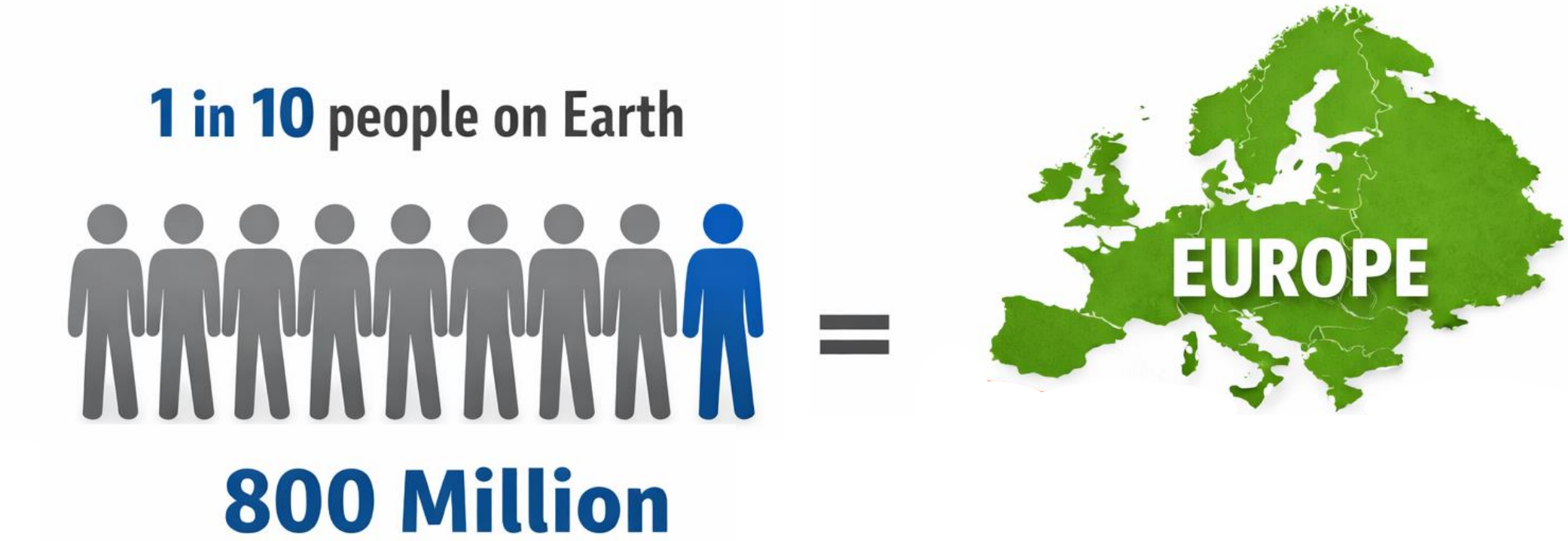
How many weekly active users does ChatGPT have?

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How many weekly active users does ChatGPT have?

2024

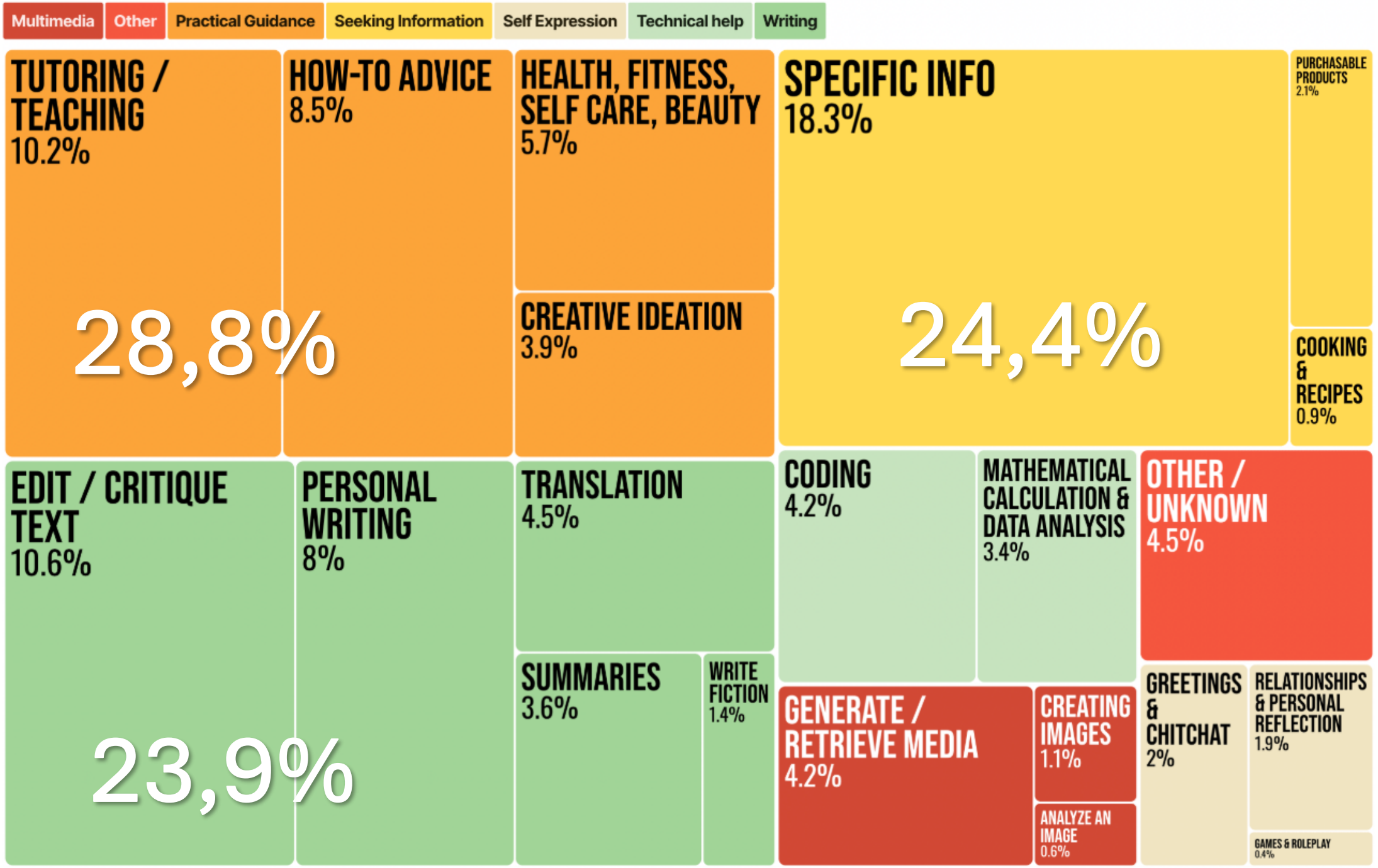


AI-generated image

ChatGPT Statistics Q3 2026: 1.2 Billion Weekly Users

The current global population is approximately 8.3 billion people as of October 2026
https://www.worldometers.info/world-population/

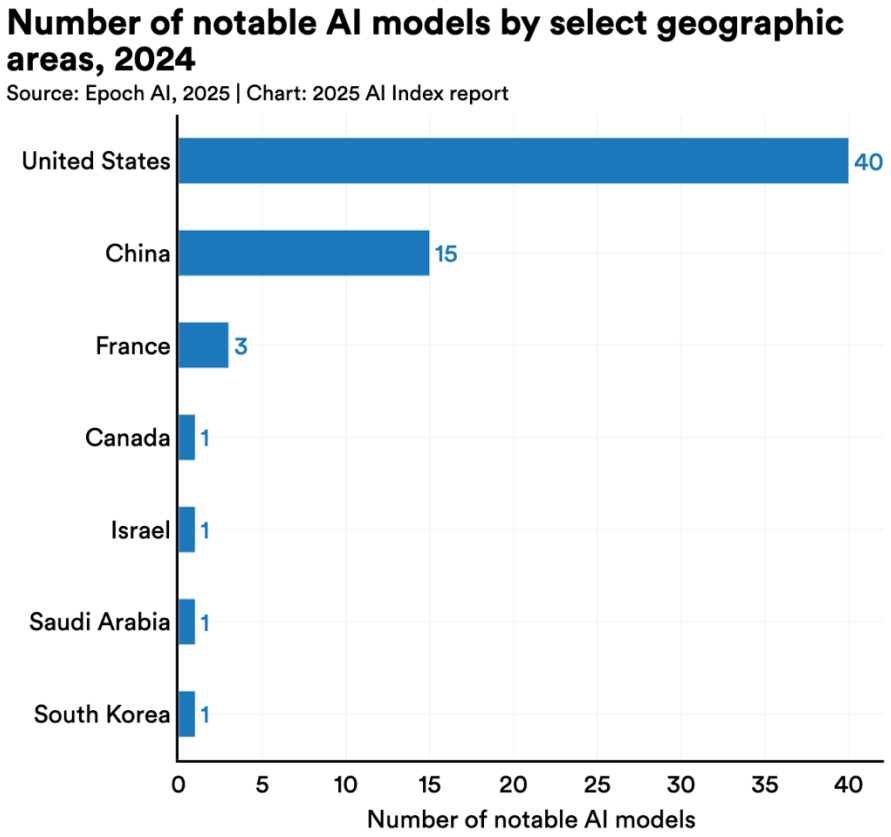
https://technologychecker.io/blog/chatgpt-statistics

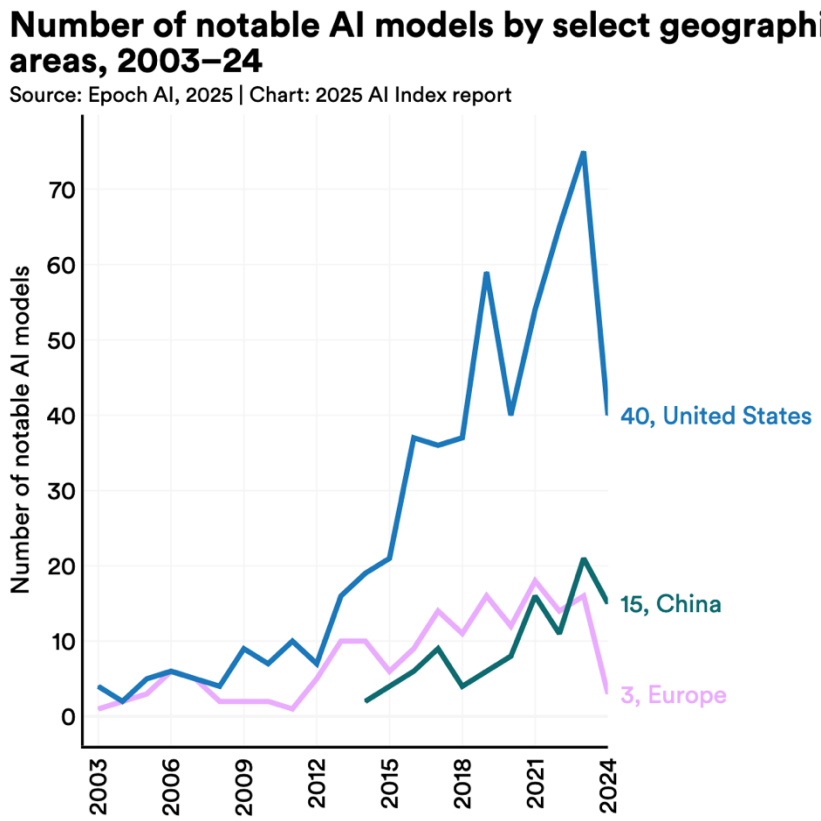


David McCandless **Information is Beautiful**

source: OpenAI

The Geographical Disparity of AI Innovation





The Cost of Training

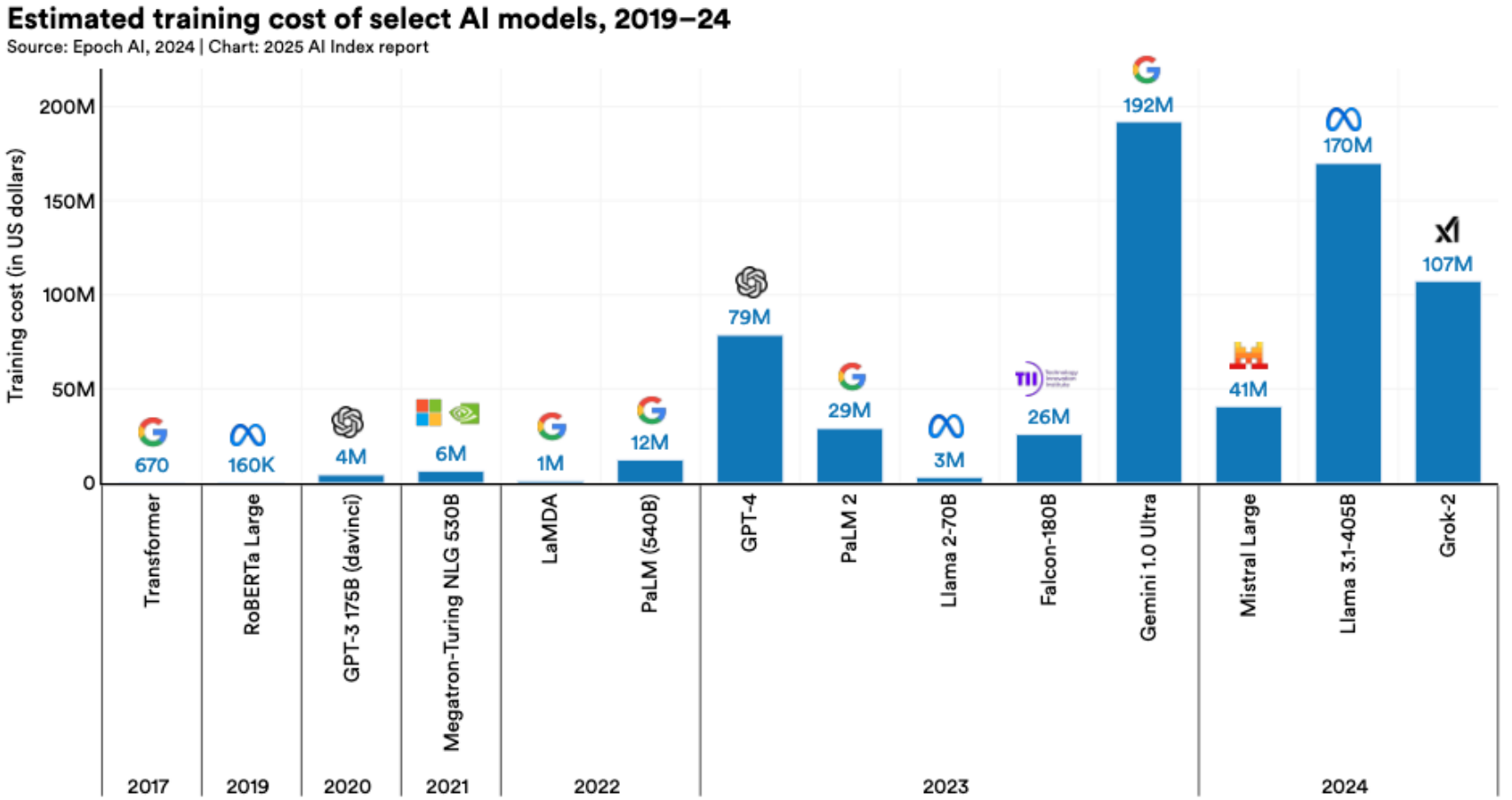


Figure 1.3.24

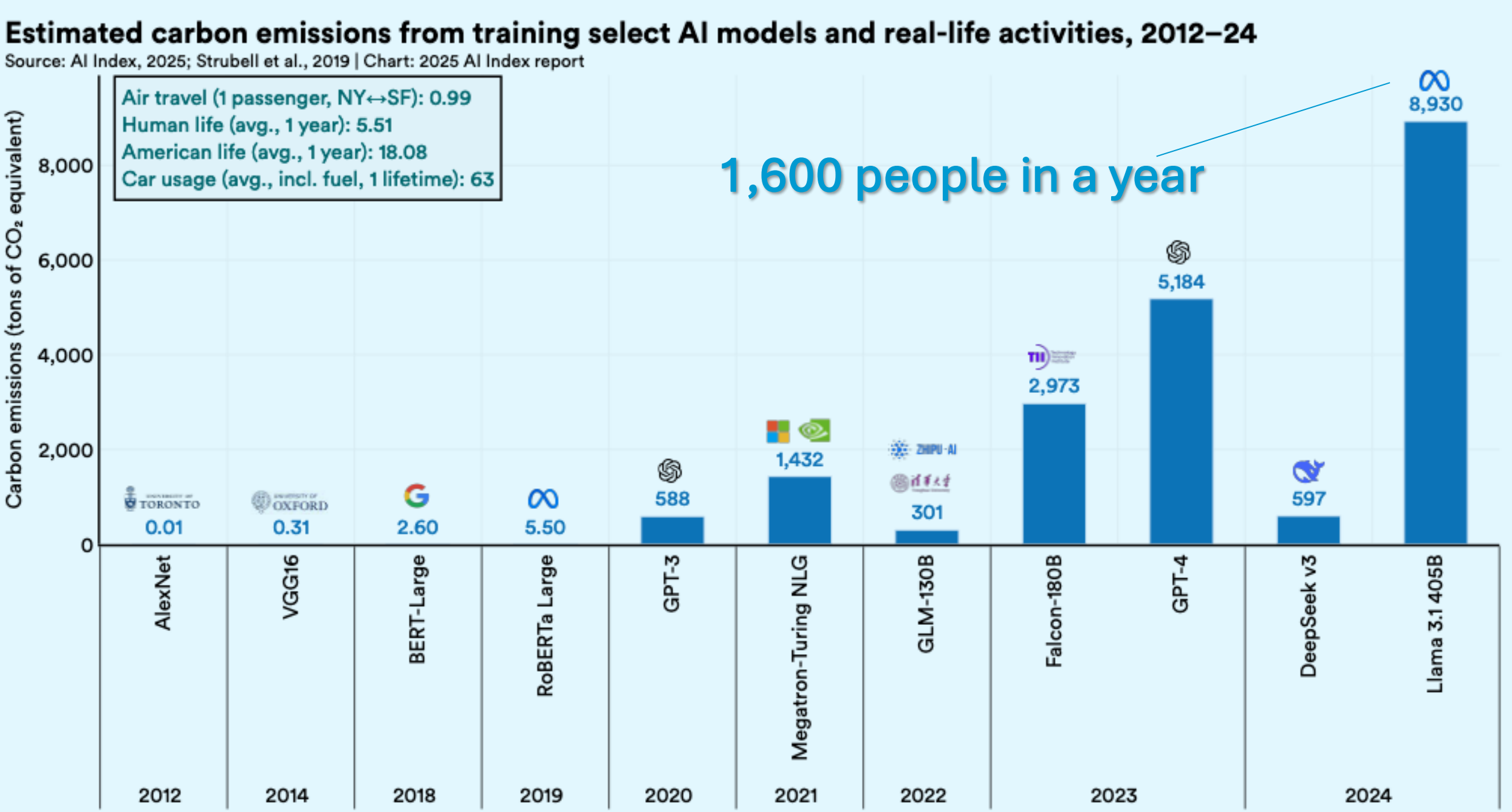

Estimated carbon emissions from training select AI models and real-life activities, 2012–24
Source: AI Index, 2025; Strubell et al., 2019 | Chart: 2025 AI Index report
Air travel (1 passenger, NY↔SF): 0.99
Human life (avg., 1 year): 5.51
American life (avg., 1 year): 18.08
Car usage (avg., incl. fuel, 1 lifetime): 63
1,600 people in a year
Carbon emissions (tons of CO₂ equivalent)
0
2,000
4,000
6,000
8,000
0.01
0.31
2.60
5.50
588
1,432
301
2,973
5,184
597
8,930
AlexNet
VGG16
BERT-Large
RoBERTa Large
GPT-3
Megatron-Turing NLG
GLM-130B
Falcon-180B
GPT-4
DeepSeek v3
Llama 3.1 405B
2012
2014
2018
2019
2020
2021
2022
2023
2024

Figure 1.4.7

All of this has been possible thanks to…

Natural Language Processing



Human language, the most **natural** to us.



Algorithms! But also Maths, coding, engineering, hardware, investments…

First, let’s try to define language

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Which language do you feel most comfortable in when speaking?

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Language

A structured system of communication that consists of grammar and vocabulary to convey meaning in spoken or signed forms

(Wikipedia)

Language

A structured system of communication → What it does

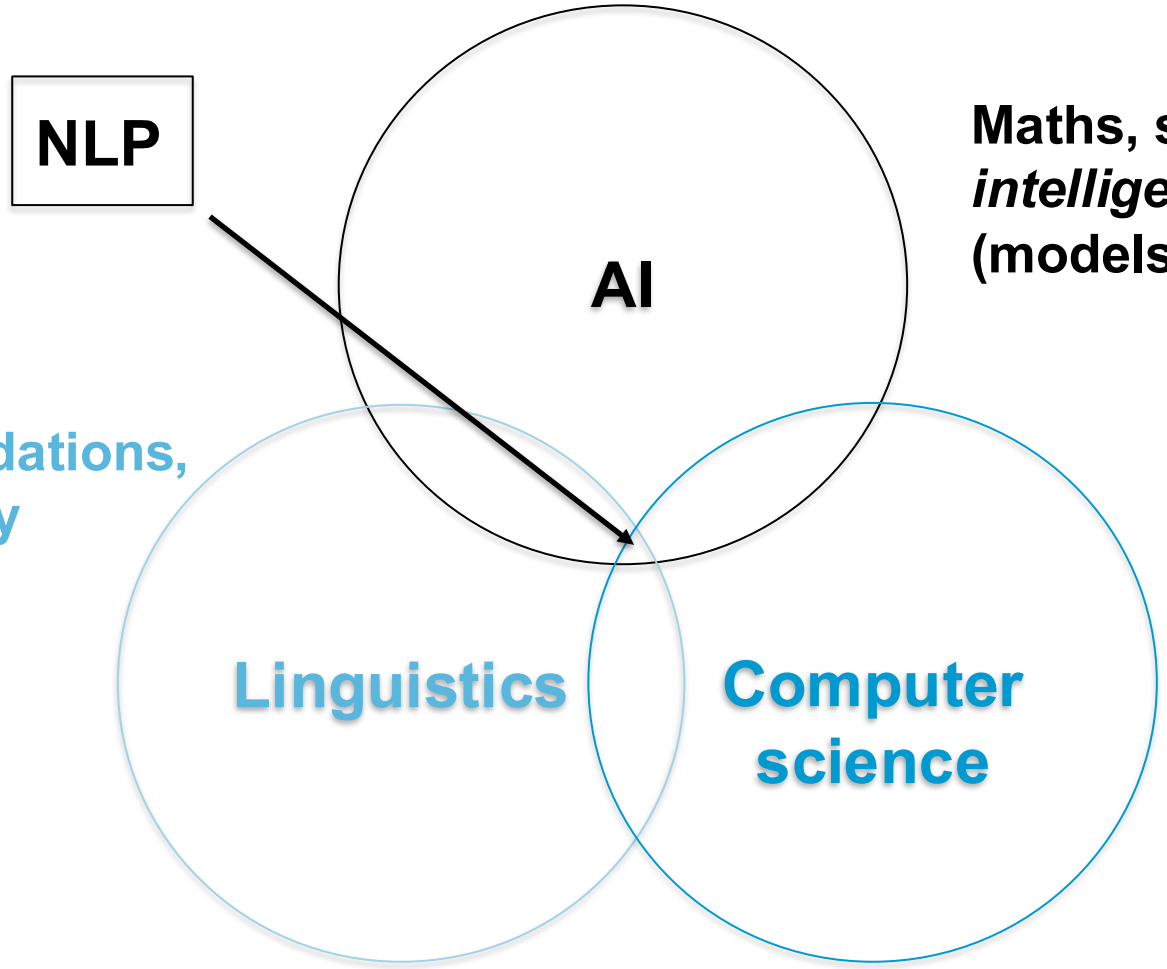
that consists of grammar and vocabulary → Entities

grammar → Rules

to convey meaning in spoken or signed forms → How

(Wikipedia)

Natural Language Processing

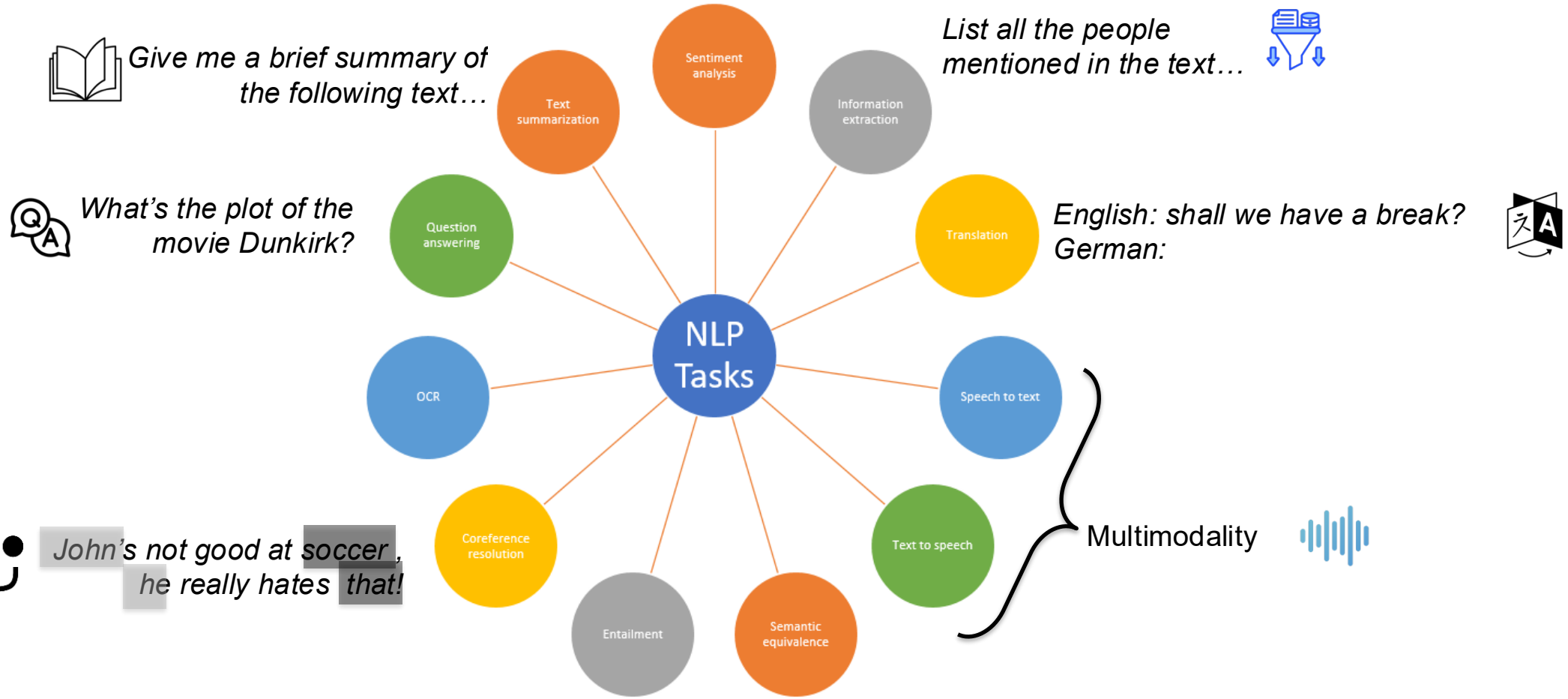


Maths, statistics, *intelligent* algorithms (models)

Theoretical foundations, history, sociology

Computation, automation, more algorithms, hardware

What could we want NLP to do?





Credit: Accelerate Cambridge

What is a large language model?

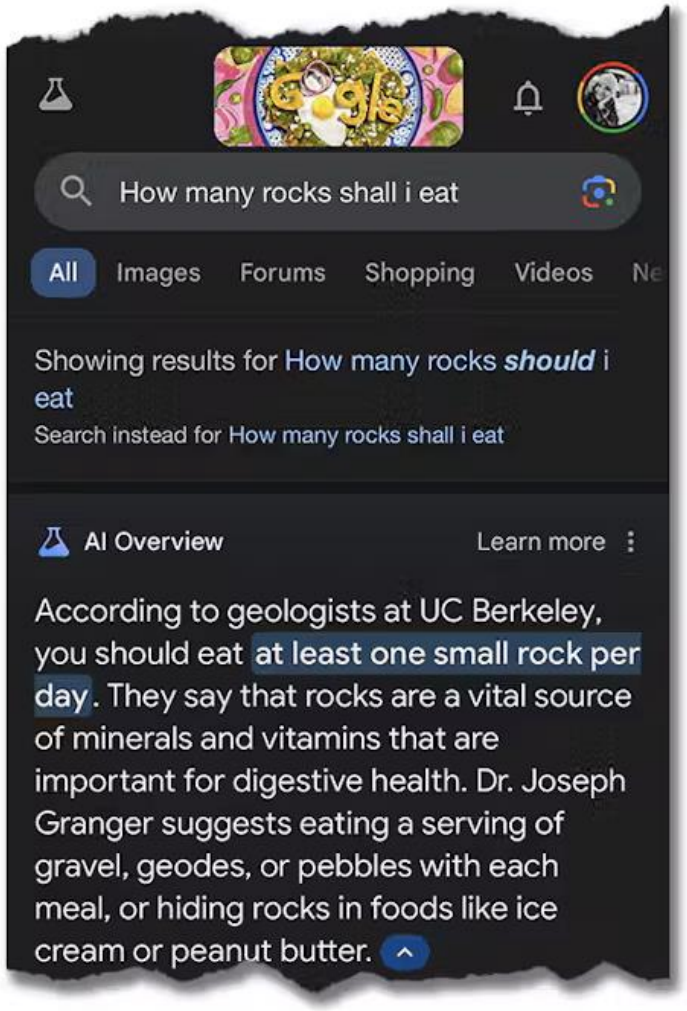
A language model is a high-parameter model that is trained on massive amounts of text with the end goal of some kind of text generation.

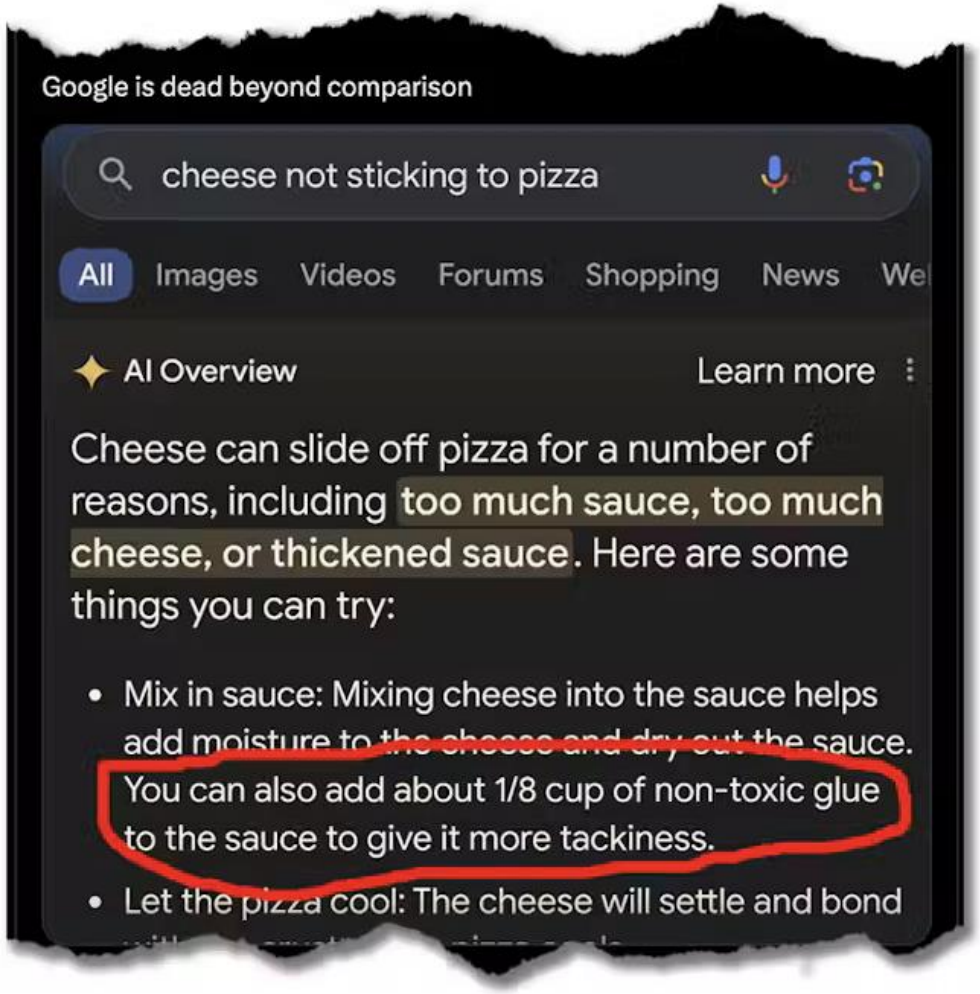
- Not all language models have the same architecture
- Not all language models are built for next-token prediction
  - GPT is a next token predictor
  - BERT is a masked-language model
  - Not all language models are chatbots

Introduction to LLMs

Some definitions …

Model The thing you train and use to generate text.
Architecture The high-level structure of the model.
Parameters (or weights) The internal knobs and dials that get altered during the course of training.
Context The input to the LLM.
Embedding A high-dimensional representation of text.
Token The numerical representation of a single piece of text.
Pretraining The first stage of training the LLM, and usually the most expensive.
Finetuning Training the LLM to do a particular task.
Inference The act of calling the model for text generation.
Quantization Reducing the size of a model.
Source The availability of the model (can be open or closed).
Modality Defined by the type of data the model is trained on (e.g. text or images).
API (Application Programming Interface) The means by which you can interact with a model.





https://www.unsw.edu.au/newsroom/news/2024/05/eat-a-rock-a-day-put-glue-on-your-pizza-how-googles-ai-is-losing-touch-with-reality

Online Harm

General Note

- This course deals with hate speech, harassment and other harmful content
- Examples on slides will be adjusted where possible
- Your projects may involve reading harmful posts: take breaks, and talk to me if it gets too much
- You can always choose a topic or dataset that you are comfortable with

- 67% of under-35s use social media for information
- 67% encountered hate speech
- LGBT+ and ethnic minorities most targeted

https://www.ipsos.com/sites/default/files/ct/news/documents/2023-11/unesco-ipsos-online-disinformation-hate-speech.pdf

Why do we care?

Harmful language is omnipresent online and it has real-world consequences:

- It has **negative mental health impacts** (e.g., depression) (Curtis et al., 2021, PNAS)
- It **discourages political commitment** (HateAid, Steinert et al., 2025, TUM Report)
- It can **fuel and amplify ongoing polarization** between groups (Brady et al., 2023, Nature Human Behavior)

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What kind of harmful content/topic could be analyzed with NLP?

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Examples

Hate Speech on Social Media

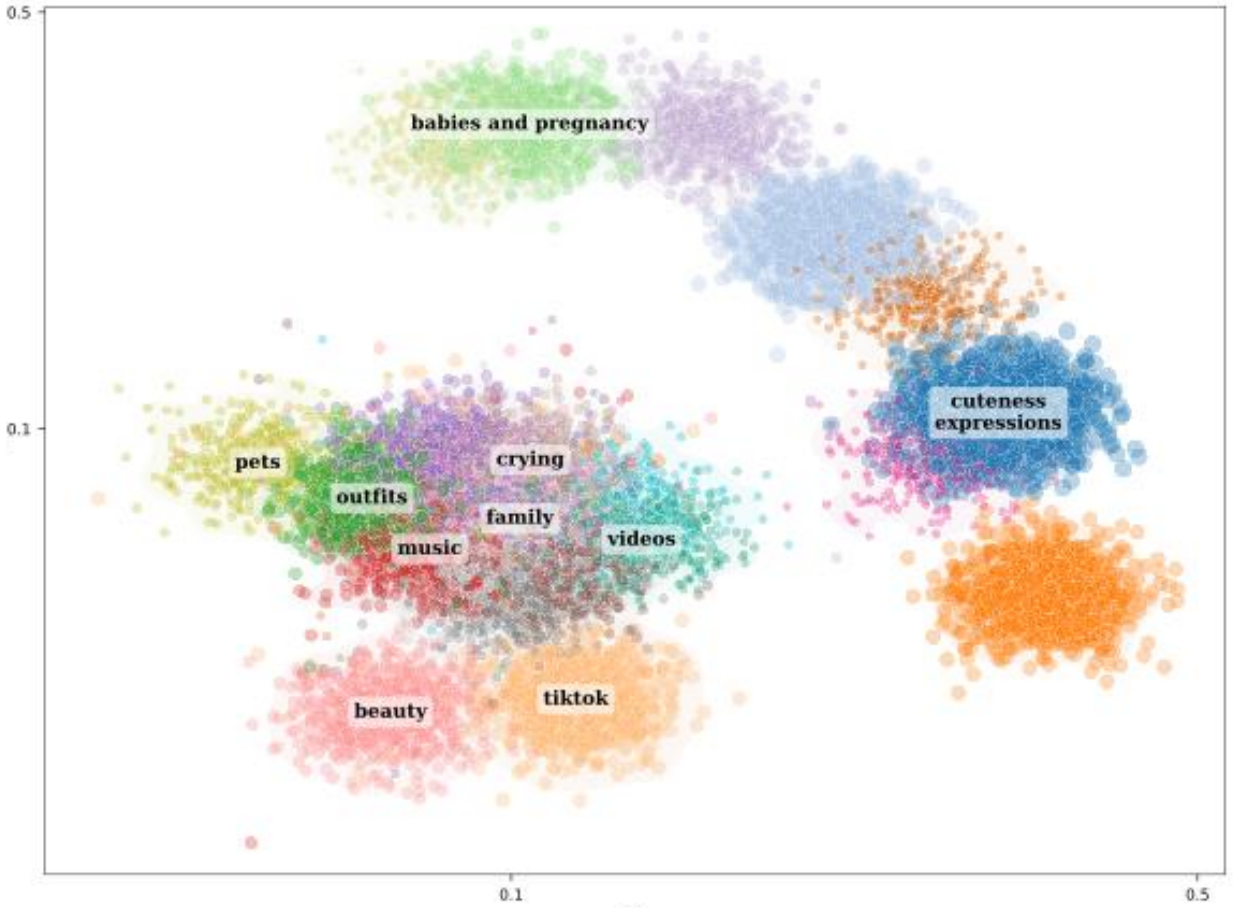
22% of the posts on an **Involuntary Celibates** forum contain **violent language**.

Matter, Schirmer, et al., 2024

Violent Experiences on Social Media

34% of the posts on the **Mental Health Subreddit** refer to **violent experiences**.

Schirmer et al., 2024



Language Context:
What do people talk about in child-related comments on TikTok?
(Schirmer et al., 2025)

What we have covered so far

What do you know about LLMs…?

What can you use them for?
What are potential problems?
What do you use them for?

What are dimensions of online harm and why do we care?

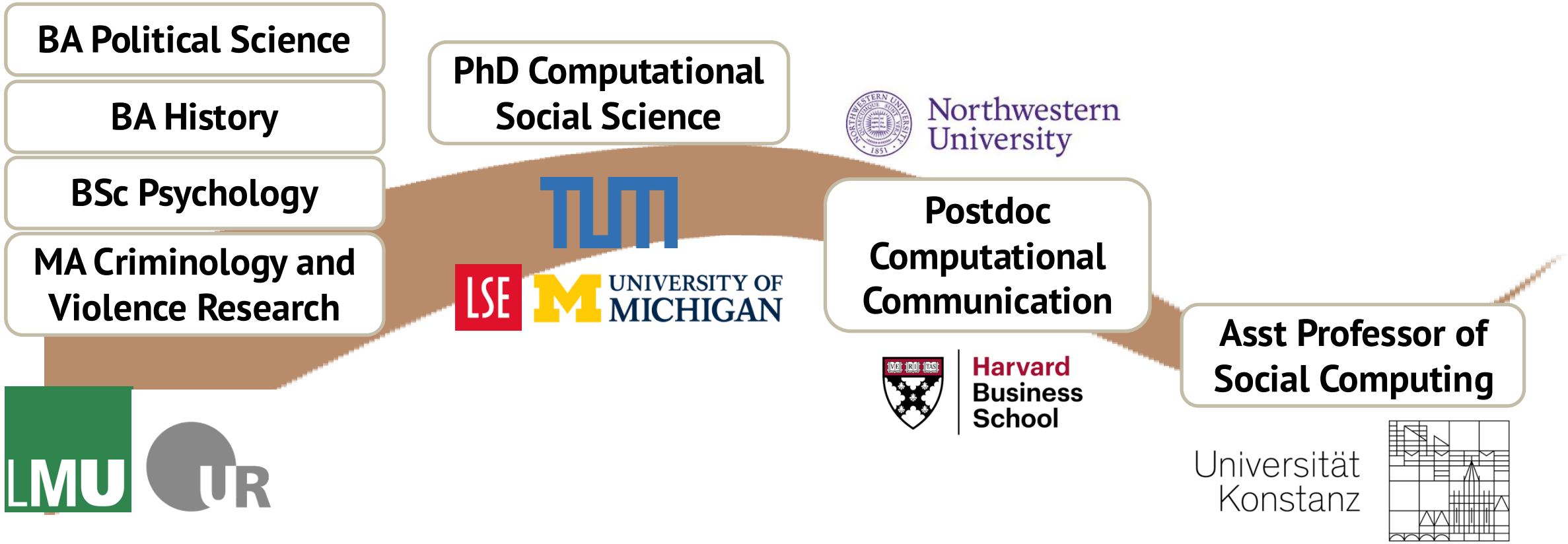
The rest of the course:

How do they work and how can we use them to build our own project?

Questions?

Who am I?

How I came to Konstanz



Research Focus

DETECTING TRAUMA IN GENOCIDE WITNESS TESTIMONIES

HATE SPEECH AND RADICALIZATION

PERCEPTIONS OF HATE AND INCITEMENT TO VIOLENCE



Your turn!

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Which program are you in?

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Have you ever written code before?

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What’s your favorite spot in Konstanz?

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This course

What you will learn

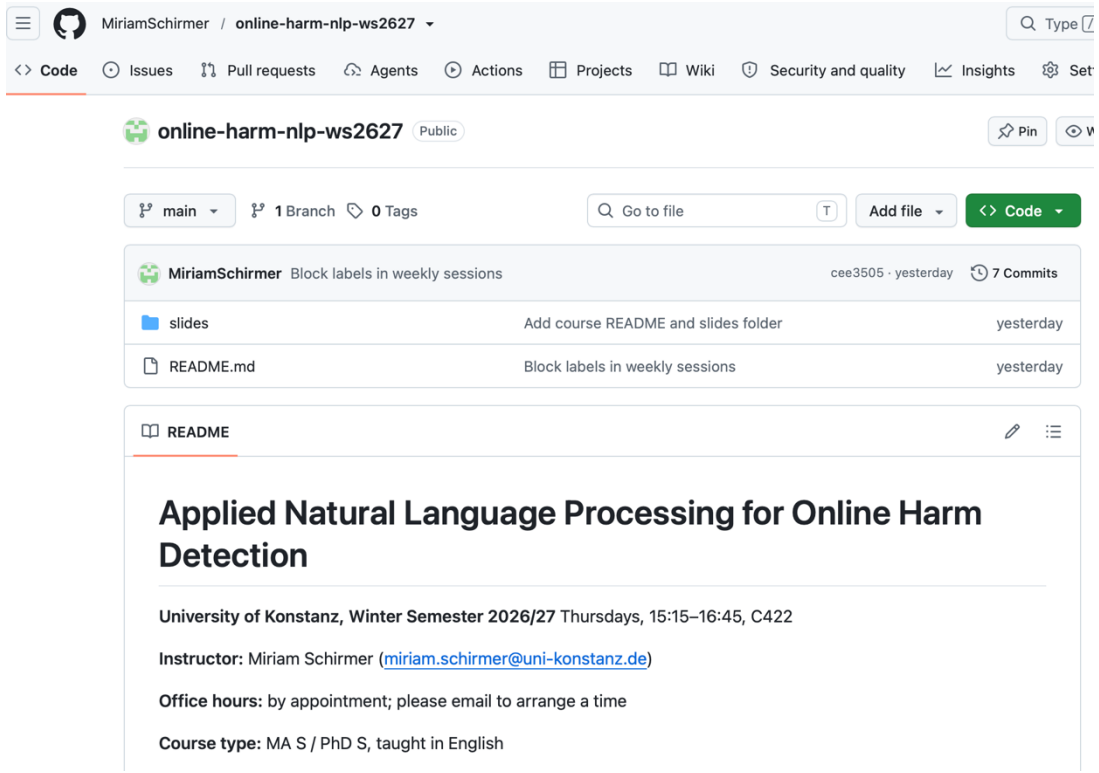
- Apply NLP methods to real-world data on harmful online discourse
- Navigate methodological and ethical challenges in this area
- Design and carry out your own small computational research project

No prior programming experience is required.

Course Structure

- **Part 1: NLP Crash Course (**sessions 2 to 4): text as data, classification, transformers and LLMs
- **Part 2: Applications** (sessions 5 to 9): three practicals on topic modeling (BERTopic), fact-checking and polarization, then mid-term presentations
- **Part 3: Advanced Topics and Projects** (sessions 10 to 14): ethics, project lab, final presentations

Schedule & Materials



Your Project

- A research question of your own on harmful online discourse, with real data (topic can be adjusted very broadly)
- Individually or in pairs (depending on course size)

- **19 Nov:** project idea (and group members) due
- **10 / 17 Dec:** mid-term presentation: question, data, planned method
- **28 Jan / 4 Feb:** final in class presentation: no final results needed!
- **Mid-February OR end of March:** final report (decide by first project presentation) **+ ~15 minutes discussion during office hours for each project separately**

Tools

- **ILIAS:** announcements – please register!
- **Google Colab:** Python in your browser, nothing to install.
- **Slido:** short questions during class, anonymous, no account needed
- **Course repository on GitHub:** notebooks and materials

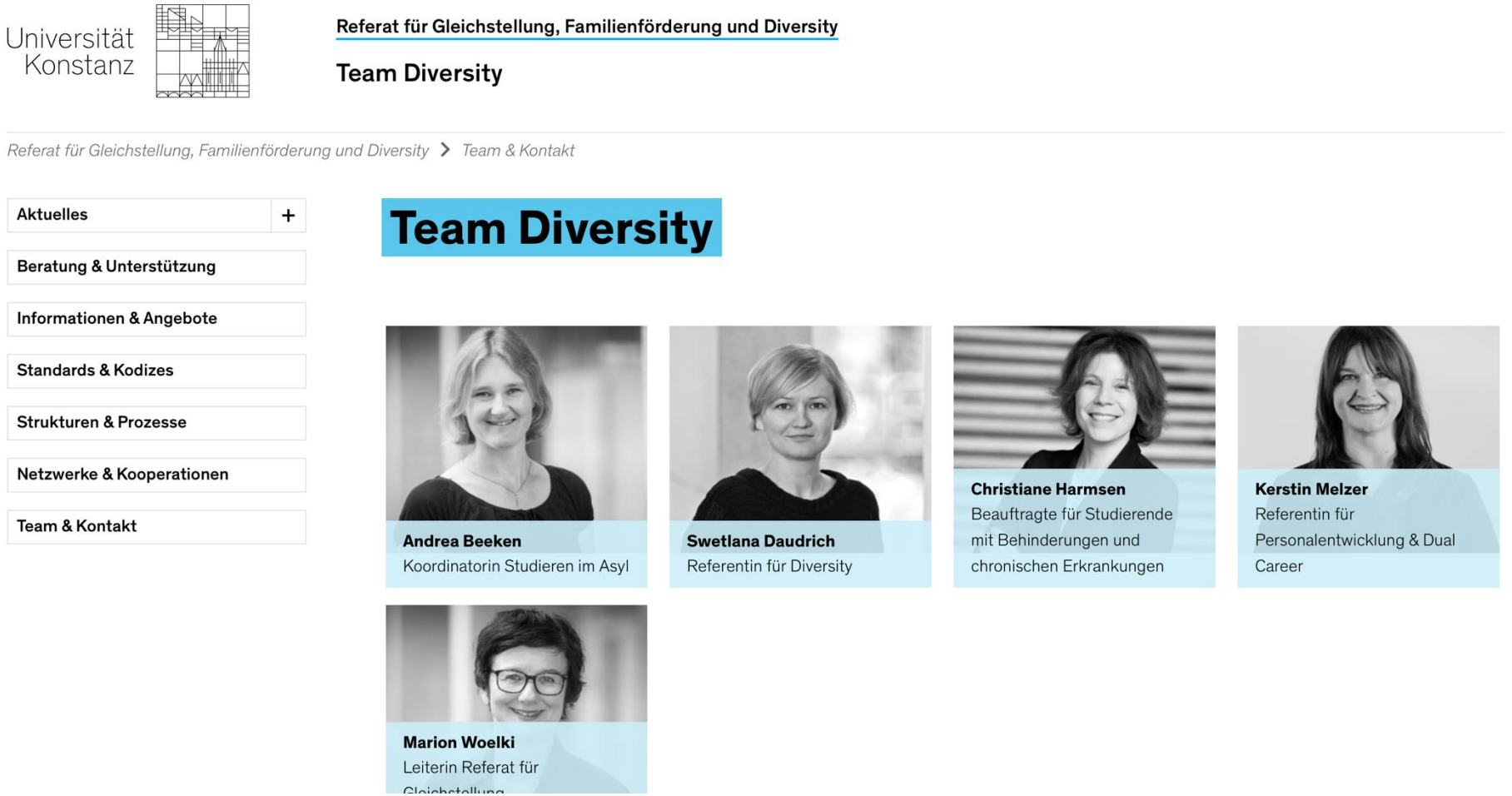
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What would you like to be able to do after this course?

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Offers the Diversity Office: https://www.uni-konstanz.de/gleichstellungsreferat/gleichstellung-in-wissenschaft-und-studium-2-1-1-1/

Before next week (29 October)

- **Readings** see github – what is one thing that you liked about one of the papers / one new thing you learned?
- **Think about** a phenomenon, a platform or a dataset you are curious about

- All readings are linked in the syllabus.

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Questions?
Thank you!

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Credits: Parts of this slide deck have been adapted from slides created by Jurafsky & Martin, Debora Nozza, and Julia Mendelsohn.